

NexaTech

Amari Bullard

University of Advancing Technology

NTW216

Prof. Jones

4/28/24

NexaTech

NexaTech is a medium-sized enterprise working towards a technological transformation. They have decided to move from physical or local hosting to cloud computing for their infrastructure. They are working with web applications, databases, and other large batch processing tasks. Given all of that, there are a few ways I would build out a cloud computing solution. First I would start off with Amazon VPC for the network setup. Amazon VPC allows you to make provisioned isolated sections of the cloud where you can use AWS resources in a virtual network that you define. This helps separate resources from other networks which is great for security reasons.

Since NexaTech is working with things like applications and databases I chose to use Amazon EC2, AWS Lambda, and Elastic load balancing to help them create a robust, efficient, and scalable environment that can handle their application needs since they are so diverse. EC2 will provide the compute capacity and overall flexibility NexaTech needs for its web applications and batch-processing tasks. We will also employ auto-scaling groups so that NexaTech can automatically adjust the number of EC2 instances they have based on the application load. This should help them save money. AWS Lambda will be used to handle some of their smaller services or event-driven processes they may have which will also help with overall cost. ELB will be the part that distributes their incoming traffic across their EC2 instances and Lambda functions. This will help with reliability and efficiency at an even greater level. These three should in theory cover their computing needs and if used properly will provide a great amount of security.

For the database services, I chose Amazon RDS, DynamoDB, and AWS Database migration service. This should be a reasonably efficient setup. Amazon RDS is there to offload

some of the challenges with database management. It'll help with that because it offers things like automated backups, patching, and scaling. This matches the needs of NexaTech quite well in my opinion. DynamoDB will provide a nosql database which should match NexaTech's need for unstructured data management. It will also help with reducing latency for web applications especially when they have high traffic. DMS will help them with efficient and reliable migration from their physical database to AWS without long amounts of downtime. It can handle many different databases.

AWS SQS is a good one to add too. SQS will allow NexaTech to separate some areas of their microservices by enabling asynchronous message queuing. This makes it so that tasks that are resource-hungry can be offloaded which improves performance and cost in some areas. This also helps with reliability.

I will use CloudWatch and Cloud Trail for their monitoring and management. These are great security tools. Cloudwatch provides in-depth monitoring which will allow NexaTech to monitor their performance on any of their services. It also can alert them if there are issues so that they can be found fast which helps security and reliability a lot. Cloud trail will help them with management. It logs all actions within an AWS account. User activities, API calls, and changes to any configurations. These are all things that need to be monitored for security reasons.

Storage would be handled with S3. This would be for storing all of their assets, backups, and their logs. I would use AWS Glacier for archiving data that isn't accessed often which will also help them with cost overall.

A solid migration plan will have to be made for them. It will depend on what they have going on with their resources in-house. Application and databases will be prioritized depending

on how important they are to avoid major downtime and do damage control. Applications will be shifted with Amazon server migration and data will be done with DMS as I stated above.

Overall I think this plan will provide them with a solid foundation for transferring to cloud computing. It will be a positive experience for them. It will save them money and enhance their security. They won't have to hire people to manage their hardware. Now their operations are also redundant in case of a disaster at their building.

References

Amazon Web Services, Inc. (n.d.). Amazon VPC user guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/toolkit-for-visual-studio/latest/user-guide/vpc-tkv.html>

Amazon Web Services, Inc. (n.d.). Amazon EC2 user guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>

GeeksforGeeks. (n.d.). Introduction to AWS Lambda. Retrieved April 28, 2024, from <https://www.geeksforgeeks.org/introduction-to-aws-lambda/>

Amazon Web Services, Inc. (n.d.). ELB: Maximizing benefits and keeping costs low. Retrieved April 28, 2024, from <https://aws.amazon.com/blogs/networking-and-content-delivery/elb-maximizing-benefits-and-keeping-costs-low/>

Amazon Web Services, Inc. (n.d.). Amazon Glacier developer guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/amazonglacier/latest/dev/introduction.html>

Amazon Web Services, Inc. (n.d.). Amazon RDS user guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Welcome.html>

Amazon Web Services, Inc. (n.d.). Create a NoSQL table. Retrieved April 28, 2024, from <https://aws.amazon.com/tutorials/create-nosql-table/>

Amazon Web Services, Inc. (n.d.). Database migration services. Retrieved April 28, 2024, from <https://aws.amazon.com/products/databases/migrations/>

Amazon Web Services, Inc. (n.d.). Amazon SQS developer guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/welcome.html>

Amazon Web Services, Inc. (n.d.). Amazon CloudWatch architecture. Retrieved April 28, 2024, from

https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/cloudwatch_architecture.html

Amazon Web Services, Inc. (n.d.). AWS CloudTrail user guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/awsccloudtrail/latest/userguide/cloudtrail-user-guide.html>

Amazon Web Services, Inc. (n.d.). Amazon S3 user guide. Retrieved April 28, 2024, from <https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>